1.

vishnu@VISHNU: ~

GNU nano 8.7.1 practical1.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[100];

    printf("Enter Linux command: ");
    scanf("%s", cmd);
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child PID : %d\n", getpid());
        execlp(cmd, cmd, NULL);
        perror("Execution Failed");
    }
    else if(pid > 0)
    {
        printf("Parent PID : %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }
    else
    {
        printf("Fork Failed\n");
    }
    return 0;
}
```

[Read 30 lines]

```
^G Help      ^O Write Out   ^F Where Is   ^K Cut     ^T Execute
^X Exit      ^R Read File   ^\ Replace    ^U Paste   ^J Justify
```

Output:

```
vishnu@VISHNU:~$ nano practical1.c
vishnu@VISHNU:~$ gcc practical1.c -o practical1
vishnu@VISHNU:~$ ./practical1.c
-bash: ./practical1.c: Permission denied
vishnu@VISHNU:~$ ./practical1
Enter Linux command: hi
Parent PID : 542
Child PID : 543
Execution Failed: No such file or directory
Child process completed.
vishnu@VISHNU:~$
```

```
vishnu@VISHNU:~$  ./practical1
Enter Linux command: uname
Parent PID : 592
Child PID : 593
Linux
Child process completed.
vishnu@VISHNU:~$  ./practical1
Enter Linux command: lscpu
Parent PID : 603
Child PID : 604
Architecture:             x86_64
  CPU op-mode(s):         32-bit, 64-bit
  Address sizes:          39 bits physical, 48 bits virtual
  Byte Order:             Little Endian
CPU(s):                   24
  On-line CPU(s) list:    0-23
Vendor ID:                GenuineIntel
  Model name:             Intel(R) Core(TM) i7-14650HX
    CPU family:           6
    Model:                183
    Thread(s) per core:   2
    Core(s) per socket:   12
    Socket(s):            1
    Stepping:             1
    BogoMIPS:             4838.39
    Flags:                fpu vme de pse tsc msr pae mce cx8 apic sep
                          r pge mca cmov pat pse36 clflush mmx fxsr s
                          se2 ss ht syscall nx pdpe1gb rdtscp lm cons
                          _tsc rep_good nopl xtopology tsc_reliable n
                          op_tsc cpuid tsc_known_freq pni pclmulqdq v
                          sse3 fma cx16 pcid sse4_1 sse4_2 x2apic mov
                          opcnt tsc_deadline_timer aes xsave avx f16c
                          and hypervisor lahf_lm abm 3dnowprefetch ss
                          brs ibpb stibp ibrs_enhanced tpr_shadow ept
                          d ept_ad fsgsbase tsc_adjust bmi1 avx2 smep
                          2 erms invpcid rdseed adx smap clflushopt c
                          sha_ni xsaveopt xsavec xgetbv1 xsaves avx_v
                          vnmi umip waitpkg gfni vaes vpclmulqdq rdpi
                          vdiri movdir64b fsrm md_clear serialize ibt
                          sh_l1d arch_capabilities
Virtualization features:
  Virtualization:         VT-x
  Hypervisor vendor:      Microsoft
  Virtualization type:    full
Caches (sum of all):
  L1d:                    576 KiB (12 instances)
  L1i:                    384 KiB (12 instances)
  L2:                     24 MiB (12 instances)
```

```
  L2:                         24 MiB (12 instances)
  L3:                         30 MiB (1 instance)
NUMA:
  NUMA node(s):               1
  NUMA node0 CPU(s):          0-23
Vulnerabilities:
  Gather data sampling:       Not affected
  Ghostwrite:                 Not affected
  Indirect target selection:  Not affected
  Itlb multihit:              Not affected
  L1tf:                       Not affected
  Mds:                        Not affected
  Meltdown:                   Not affected
  Mmio stale data:            Not affected
  Old microcode:              Not affected
  Reg file data sampling:     Mitigation; Clear Register File
  Retbleed:                   Mitigation; Enhanced IBRS
  Spec rstack overflow:       Not affected
  Spec store bypass:          Mitigation; Speculative Store Bypass disabl
                              ia prctl
  Spectre v1:                 Mitigation; usercopy/swapgs barriers and __
                               pointer sanitization
  Spectre v2:                 Mitigation; Enhanced / Automatic IBRS; IBPB
                              ditional; PBRSB-eIBRS SW sequence; BHI BHI_
                              S
  Srbds:                      Not affected
  Tsa:                        Not affected
  Tsx async abort:            Not affected
  Vmscape:                    Not affected
Child process completed.
vishnu@VISHNU:~$  ./practical1
Enter Linux command: ps
Parent PID : 618
Child PID : 619
    PID TTY          TIME CMD
    357 pts/0    00:00:00 bash
    618 pts/0    00:00:00 practical1
    619 pts/0    00:00:00 ps
Child process completed.
vishnu@VISHNU:~$  ./practical1
Enter Linux command: top
Parent PID : 631
Child PID : 632
top - 16:42:27 up 6 min,  1 user,  load average: 0.02, 0.05, 0.01
Tasks:  26 total,   1 running,  25 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,
MiB Mem :  11773.7 total,  10488.0 free,    658.4 used,    799.1 buff/ca
MiB Swap:   3072.0 total,   3072.0 free,      0.0 used.  11115.3 avail M
```

2.

GNU nano 8.7.1

```c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int src, dest;
    char buffer[1024];
    int bytes;

    src = open("input.txt", O_RDONLY);

    if(src < 0)
    {
        printf("Cannot open source file\n");
        return 1;
    }

    dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    while((bytes = read(src, buffer, sizeof(buffer))) > 0)
    {
        write(dest, buffer, bytes);
    }

    close(src);
    close(dest);

    printf("File copied successfully.\n");

    return 0;
}
```

output:

```
vishnu@VISHNU:~$ nano practical1a
vishnu@VISHNU:~$ gcc practical1a.c -o practical1a
vishnu@VISHNU:~$ ./practical1a
Cannot open source file
vishnu@VISHNU:~$ ./practical1a
Cannot open source file
vishnu@VISHNU:~$ nano practical1a
vishnu@VISHNU:~$ nano practical1a.c
vishnu@VISHNU:~$ gcc practical1a.c -o practical1a
vishnu@VISHNU:~$ ./practical1a
Cannot open source file
vishnu@VISHNU:~$ nano practcal1a.c
vishnu@VISHNU:~$ nano input.txt
vishnu@VISHNU:~$  ./practical1a
File copied successfully.
vishnu@VISHNU:~$  ./practical1a
File copied successfully.
vishnu@VISHNU:~$
```